

Fake Account Detection

Using Artificial Neural Networks (ANN)

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Executive Summary

Automating Threat Detection

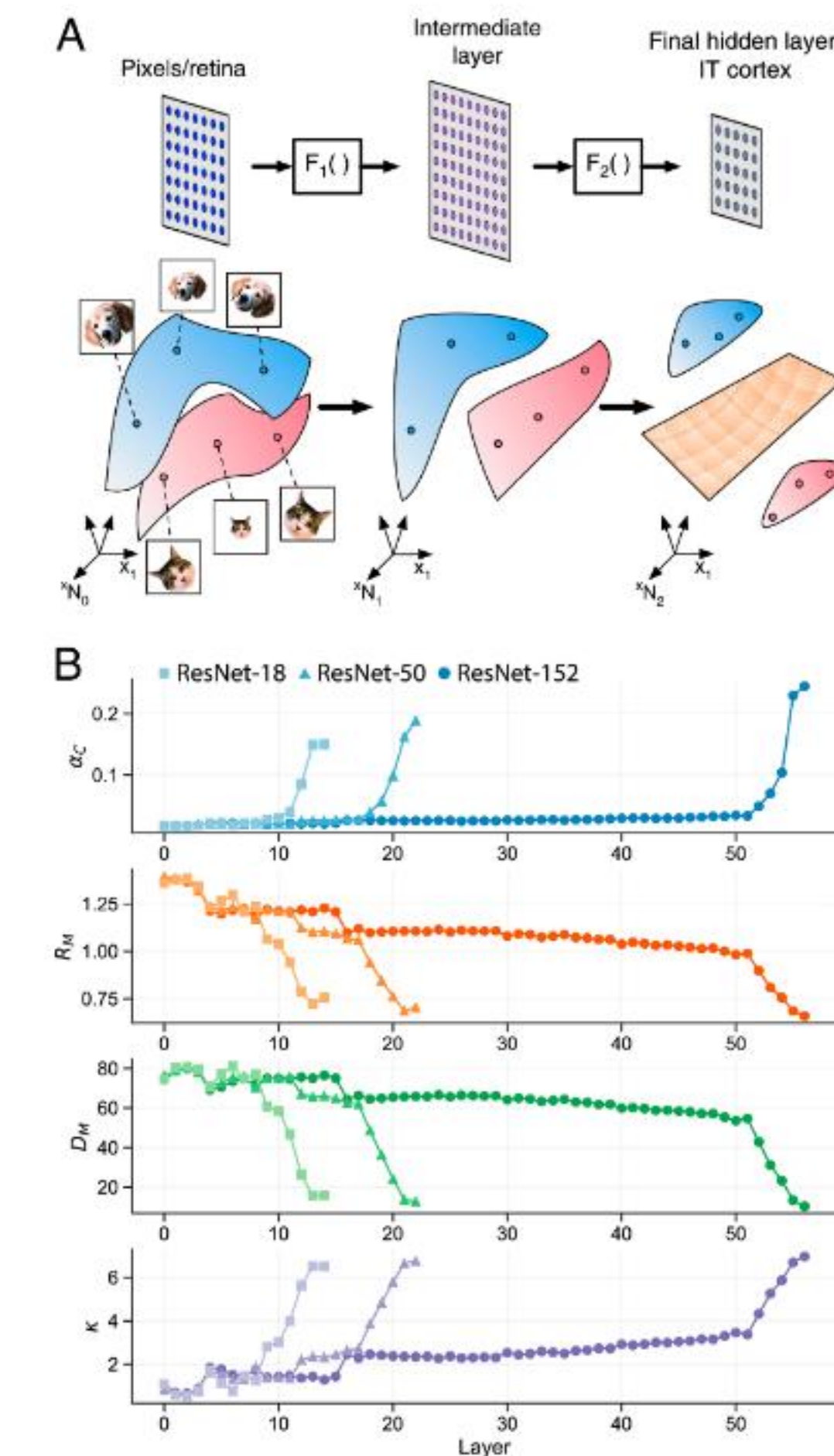
Social media platforms face significant challenges in identifying fake accounts due to the increasing sophistication of automated bots and malicious users. Traditional rule-based systems often fail to detect evolving fraudulent patterns.

This project leverages Deep Learning techniques to build an intelligent classification model capable of distinguishing fake accounts from legitimate users, improving efficiency and reducing manual verification efforts.



Problem Statement

- Fake social media accounts have become a major cybersecurity threat across various online platforms.
- They are commonly used for phishing attacks, spam campaigns, misinformation dissemination, identity theft, and financial fraud.
- Manual identification of fake accounts is time-consuming, costly, and impractical due to massive user volumes.
- Our objective is to develop an automated Artificial Neural Network (ANN) to accurately classify these accounts.



Data Sources & Key Features

Target & Scope

The dataset contains profile-related features and behavioral attributes to identify suspicious behavior.

- **Dataset:** Fake Social Media Account Detection Dataset (CSV format).
- **Target Variable:** is_fake (1: Fake Account, 0: Genuine Account).

Key Indicators

- Platform, Profile Picture, Bio Length
- Followers, Following, Follower-Following Ratio
- Account Age (Days), Number of Posts, Posts Per Day
- Caption & Content Similarity Scores
- Spam & Generic Comment Rates

Dataset Overview

3000

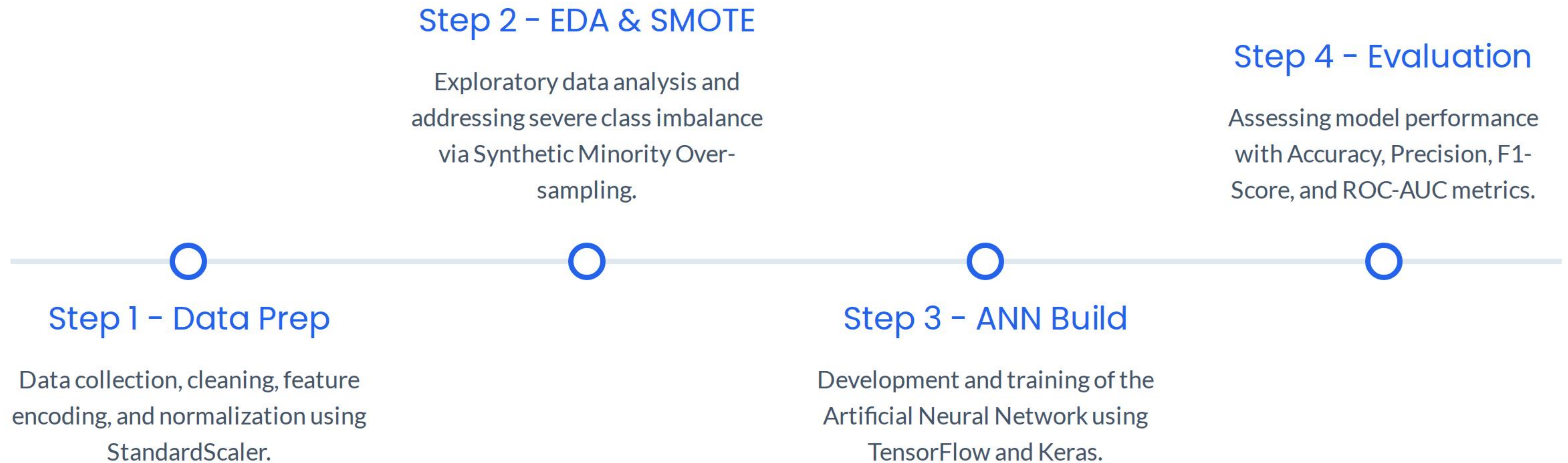
Total Records Analyzed

Highly Imbalanced Data

The initial exploratory data analysis (EDA) revealed a significant class imbalance in the target variable, which is typical for fraud detection datasets.

- **Fake Accounts:** 2,993 instances.
- **Genuine Accounts:** 7 instances.
- **Features:** 18 comprehensive attributes.
- **Data Types:** 4 floats, 13 integers, 1 object.
- **Quality:** Zero missing values, zero duplicates.

Project Methodology



Data Preprocessing & EDA



Cleaning

Verified dataset integrity by confirming zero missing values and zero duplicate records, ensuring a solid foundation for model training.



Encoding

Transformed the 'platform' column, the sole categorical feature, into numerical format using LabelEncoder for neural network compatibility.



Scaling

Applied StandardScaler to normalize all numerical features, bringing them to a similar range to optimize gradient descent and convergence.

Handling Class Imbalance

50/50
Perfect Class Balance

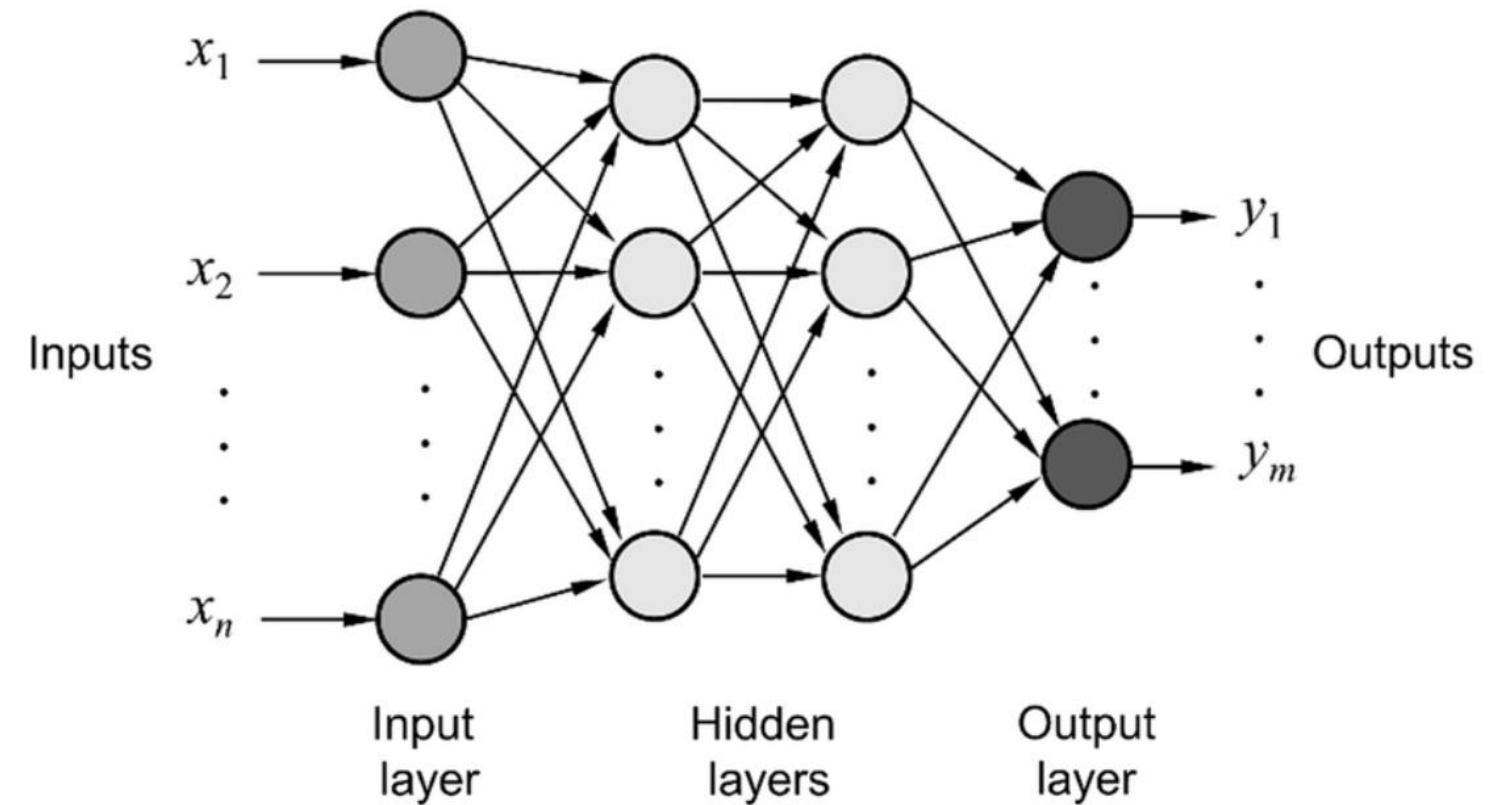
SMOTE Implementation

The dataset contained significantly more fake accounts than genuine ones, which can lead to biased predictions. We utilized the Synthetic Minority Over-sampling Technique (SMOTE).

- **Before SMOTE:** 2993 Fake vs. 7 Genuine.
- **Process:** Generated synthetic samples for the minority class to create a balanced dataset.
- **After SMOTE:** 2993 Fake vs. 2993 Genuine.
- **Train/Test Split:** Data split 80/20 with stratification to maintain this balance.

ANN Model Architecture

- **Input Layer:** 128 neurons utilizing ReLU activation, followed by Batch Normalization and 30% Dropout to prevent overfitting.
- **Hidden Layer 1:** 64 neurons with ReLU, Batch Normalization, and 30% Dropout.
- **Hidden Layer 2:** 32 neurons with ReLU activation.
- **Hidden Layer 3:** 16 neurons with ReLU activation, progressively extracting high-level features.
- **Output Layer:** 1 neuron with Sigmoid activation for robust binary classification (Fake vs. Genuine).



Model Training & Config

Compilation Parameters

The model was compiled to optimize specifically for binary classification tasks.

- **Optimizer:** Adam (Adaptive Moment Estimation) for efficient gradient updates.
- **Loss Function:** Binary Crossentropy.
- **Metrics:** Accuracy.

Training Strategy

Training was structured to maximize learning while rigorously preventing overfitting.

- **Batch Size:** 32, ensuring stable weight updates.
- **Epochs:** Up to 50 iterations.
- **Validation Split:** 20% of training data.
- **Early Stopping:** Implemented with a patience of 5 epochs, restoring the best model weights.

Model Evaluation Metrics

100%

Test Accuracy Achieved

Flawless Classification

The trained Artificial Neural Network performed exceptionally well on the unseen 20% test dataset (1,198 samples).

- **Test Loss:** 0.0000226 (Extremely low error rate).
- **Precision:** 1.00 for both Fake and Genuine classes.
- **Recall:** 1.00 for both classes, indicating no missed detections.
- **F1-Score:** 1.00, demonstrating a perfect balance between precision and recall.

Confusion Matrix & ROC-AUC

Perfect Discrimination

The model's Confusion Matrix confirms zero false positives and zero false negatives across the entire test set.

- ✓ **True Positives:** 599 Fake accounts correctly identified.
- ✓ **True Negatives:** 599 Genuine accounts correctly identified.
- 📈 **ROC-AUC Score: 1.0.** The Receiver Operating Characteristic curve shows a perfect Area Under the Curve, signifying flawless class separation capability.

—————→ **Data wrangling,
transformation, and analysis** —————



Data visualization desert



Risks and Challenges

-  **Class Imbalance:** The raw data was highly skewed. While SMOTE successfully mitigated this, synthetic data may not capture all nuanced real-world edge cases of genuine accounts.
-  **Overfitting Risks:** Deep learning models are prone to memorizing training data. We addressed this using robust Dropout layers and Early Stopping mechanisms during training.
-  **Computational Requirements:** ANNs require greater computational power and extended training times compared to traditional machine learning algorithms.
-  **Model Generalization:** The model performs perfectly on the current dataset, but evolving bot behaviors and new platform structures will require continuous retraining and feature updates.

Expected Outcomes & Impact

 Person reviewing cyber security alerts

Automated Identification: Successfully developed a DL model that accurately classifies fake accounts, minimizing manual review time.



Behavioral Insights: Improved detection by leveraging a combination of profile metrics and complex user activity patterns.



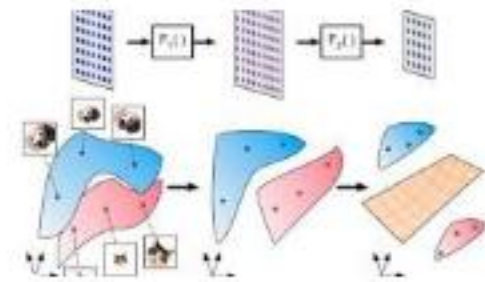
Enhanced Cybersecurity: Provides a scalable foundation for social media platforms to deploy real-time fraud detection and protect user trust.

Image Sources



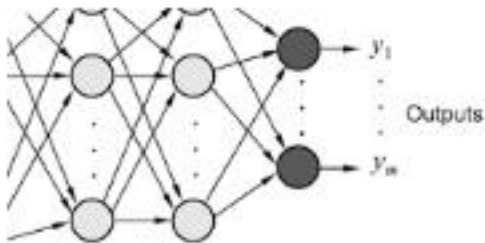
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